

DETERMINATION OF METHANE SOURCES GLOBALLY BY SCIAMACHY

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INTRODUCTION

- Looking at the long-term trend, the global average temperature **increased by 0.74 ± 0.18 °C** over the 100-year period ending in 2005 (IPCC, 2007).
- Much research has found that the main cause of global warming is increased greenhouse gases in the atmosphere due to the use of **fossil fuels by humans** and is not a natural phenomenon (IPCC, 2007; Houweling et al., 2000).
- Greenhouse gases are materials in the atmosphere that **absorb and re-emit** some of the infrared light radiated from the surface of the earth.
- Carbon dioxide (CO₂), methane (CH₄), dinitrogen monoxide (N₂O) and chlorofluorocarbon gases are all greenhouse gases.

INTRODUCTION

- The global warming potential (**GWP**) is a numeric indicator that represents how much a particular greenhouse gas contributes to global warming.
- Across 100 years, **CH₄ has a global warming effect of 25** (that is, 25-fold the effect of CO₂), N₂O has an effect of 310.
- Comparing the concentration levels present before the industrial revolution with the mean concentrations in 2012, CO₂ concentrations have increased from 278 ppm to 393.1 ppm (a 41% increase), **CH₄ has increased from 700 ppb to 1819 (160%)**, and N₂O has increased from 270 ppb to 325 ppb (20%).
- The increases in the greenhouse gases by the GWP are 63% for CO₂, **18% for CH₄**, 6% for N₂O, and 13% for other (IPCC, 2007)

Satellite measurement

- **SCIAMACHY**

- Scanning Imaging Absorption Spectrometer for Atmospheric Chartography (SCIAMACHY) sensor mounted on the Environmental Satellite (ENVISAT), which was launched on 1 March 2002 by the European Space Agency (ESA)

- **GOSAT**

- Greenhouse Gases Observing Satellite (GOSAT), which was launched on 23 January 2009 through cooperation between the Japan Aerospace Exploration Agency (JAXA), the National Institute for Environmental Studies (NIES) of Japan, and the Ministry of the Environment (MOE) of Japan.

- Although the greenhouse gas data measured by the WDCGG cover only a narrow range of geographical locations relative to the globe, satellite collection allows measurement over **two-dimensional areas** and is often used in research **covering wide areas**

Sources of CH₄

- Sources of CH₄ are broadly divided into non-biological sources and biological sources.
 - Non-biological sources of CH₄ emissions include mining and burning of fossil fuels, burning of biomass, and garbage disposal.
 - Biological sources include swamps, wet rice paddies, livestock, landfill, termites, and forests.
 - In the processes of biological ecosystems, CO₂ is produced when organic matter is decomposed by anaerobic bacteria, and CH₄ is produced by an abundance of CO₂ and methane bacteria (Conrad, 1996).
- Biological sources account for over 70% of total CH₄ emissions (IPCC, 2007).

Emitters of CH₄

- Emitters of CH₄ include natural and artificial sources.
 - **Natural sources**, the largest sources of emissions are bogs and swamps, with emissions also occurring from termites and vegetation.
 - **Artificial sources**, the main sources are wet rice paddies, livestock, waste disposal processes, and mining and use of fossil fuels
- The amount of CH₄ that is emitted from sources such as **swamps and wet rice paddies** has been clarified by **many field observations**
- However, the amount of CH₄ emitted varies according to differences in the observation targets and environment.

Sinks for CH₄

- Sinks for CH₄, including absorption by **soil (5%)**, decomposition by **chemical reaction** in the atmosphere accounts for over **87%** of CH₄ elimination.
- Decomposition by chemical reaction occurs via reaction with hydroxide radicals (**OH radicals**) by means of chemical reactions involving sunlight, oxygen, ozone, water vapor, etc.

Objectives

- In this research, we estimated the surface concentration of methane emissions and extracted the CH₄ sources by using time-series data from [SCIAMACHY](#) covering the period from 2003 to 2011.
- We treat the surface background CH₄ concentration, which has been problematic in other research, as the CH₄ concentration over ocean at the same latitude, and then calculate the surface concentration of CH₄ emissions under this assumption.
- Furthermore, we compare the topographical features of CH₄ sources with vegetation maps to investigate the sources of CH₄ emissions.

DATA

- **SCIAMACHY**

- SCIAMACHY was mounted on ENVISAT and operated from February 2002 to April 2012. The main aim of SCIAMACHY was to measure the various greenhouse gases and atmospheric pollutants in the troposphere and stratosphere.
- The vertical column density (VCD) data for CH₄ are retrievable by using the iterative maximum a posteriori (IMAP)-DOAS method of Frankenberg et al. (2005).

DATA

- **GLCMNO**

- The Global Land Cover by National Mapping Organizations (GLCNMO) is available as a cluster dataset divided into 20 items according to the land cover classification system (LCCS) developed by the Food and Agriculture Organization of the United Nations (FAO).
- The landcover legend includes evergreen broad-leaf forest, deciduous broad-leaf forest, evergreen needle-leaf forest, deciduous needle-leaf forest, mixed forest, grassland, fields, [wet rice paddies](#), mangroves, swamps, bare ground (sand), and bare ground (pebbles and stone) (Tateishi et al., 2014).

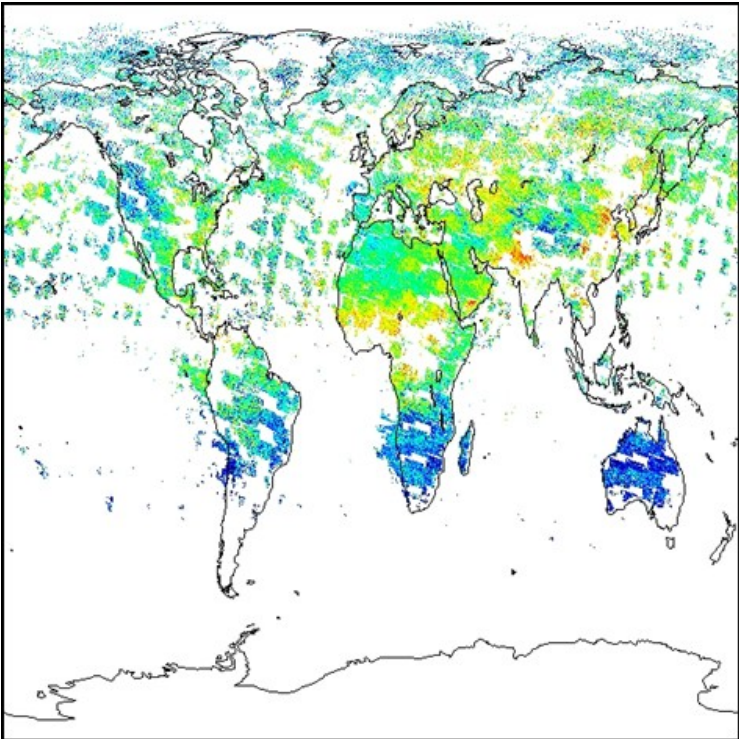
WDCGG

- Observation of CO₂ as a greenhouse gas began in 1957 at the South Pole and in 1958 at Mauna Loa, Hawaii.
- Measurements of greenhouse gases have been conducted by the World Data Center for Greenhouse Gases (WDCGG).
- The WDCGG was established under the Global Atmosphere Watch (GAW) process of the World Meteorological Organization (WMO), and the WDCGG **collects, manages, and provides** data on gases related to greenhouse gases (CO₂, CH₄, CFCs, N₂O) measured in the atmosphere and ocean.

Data Process

- Cloud-free Images

- Optical sensor data obtained from satellites are often unclear near the ground surface due to the effects of clouds.
- We used the threshold data (selection_criteria.txt) from the SCIAMACHY product site (ftp://ftp.sron.nl/pub/pub/DataProducts/SCIAMACHY_CH4/)



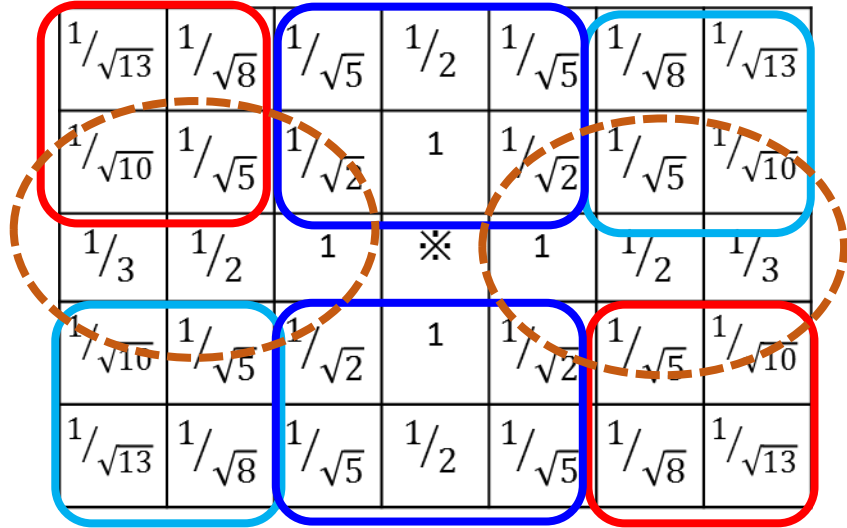
When a large proportion of data is missing due to the effects of cloud, analysis becomes difficult.



Spatial and Time-series interpolation to impute missing data

Spatial Interpolation

weighted spatial filter



is missing Data

$$T(x, y) = \sum_{j=-2}^2 \sum_{i=-3}^3 f(x+i, y+j) W(i+3, j+2)$$

+ W

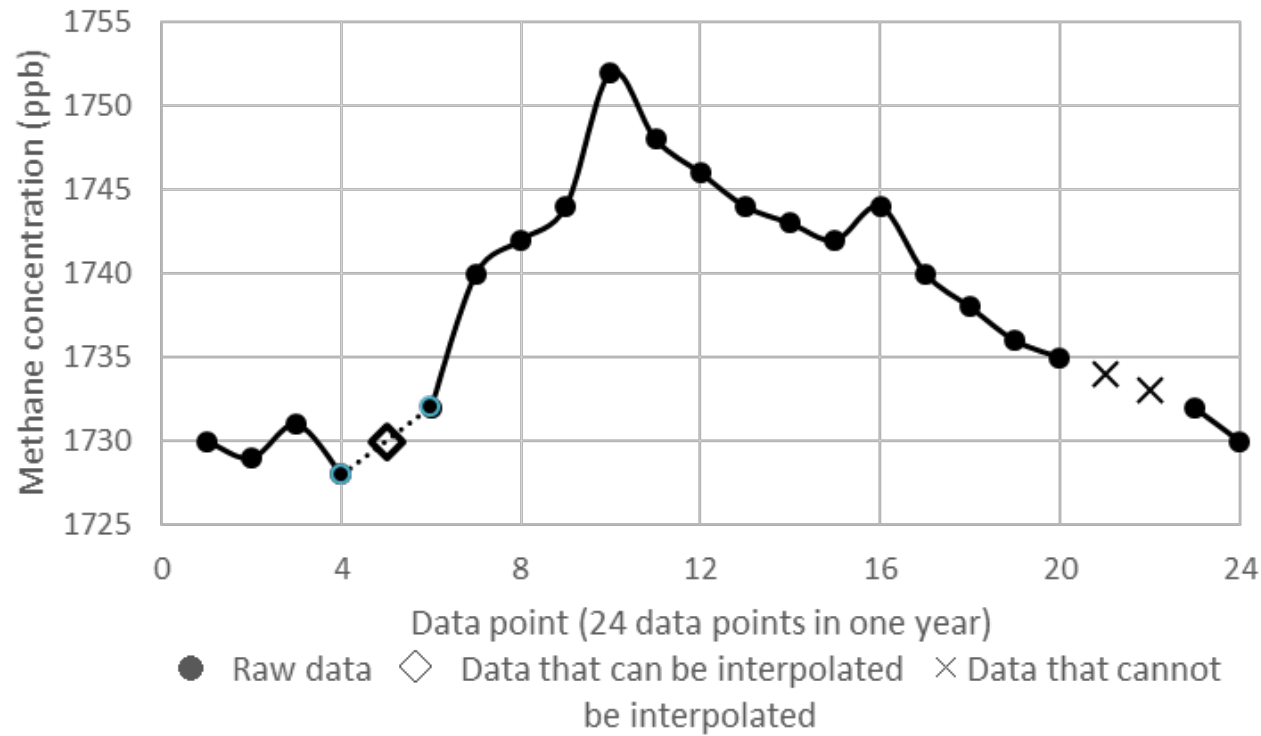
$$f'(x, y) = T(x, y) / \text{Sum of Weight}$$

If target data is missing,

1. Condition for spatial interpolation

- First, we divided the 34 pixels surrounding the target pixel into **8 regions**, and if pixels without cloud existed in **at least one pair of opposite directions** (0° and 180°, 45° and 225°, 90° and 270°, 135° and 315°), the value for the target pixel was interpolated by applying the **7 × 5 weighted filter**

Time-series interpolation

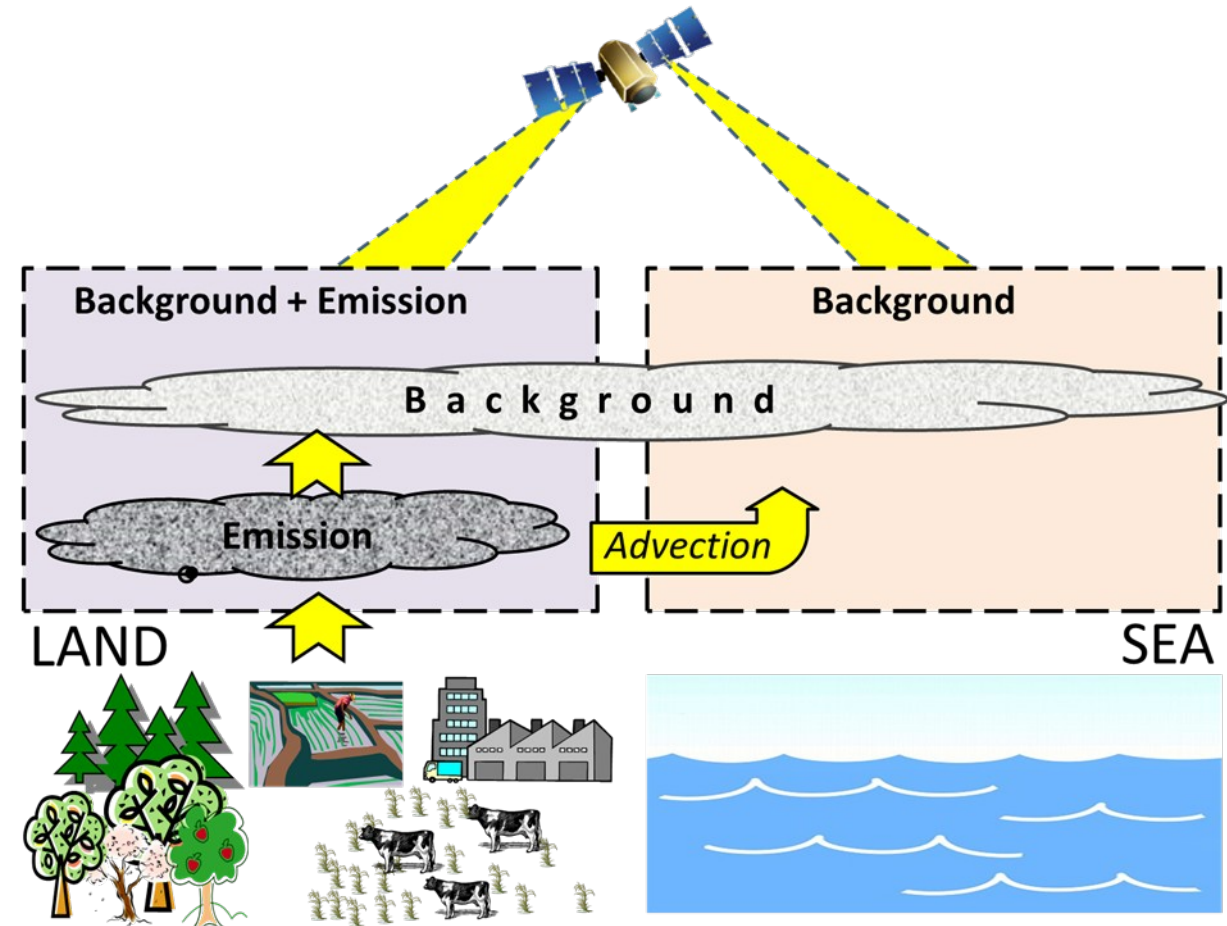


The fifth data point is missing, linear interpolation is performed using the fourth (1728) and sixth (1732) data points. However, for data points 21 and 22, interpolation is not performed because consecutive data points are missing.

- A necessary condition for time-series interpolation of CH_4 concentration in the atmosphere was that the CH_4 concentration did not change sharply over a short period of time (here, less than 30 days).
- if a target pixel was missing
- the value was found by linear interpolation using the data before (15 days before) and after (15 days after) the target pixel

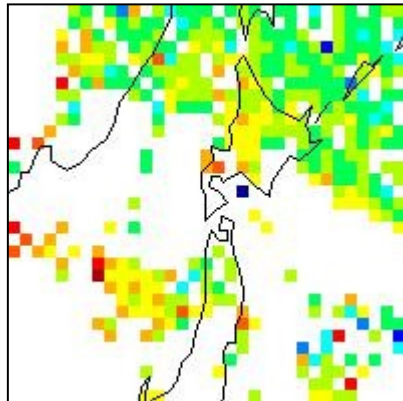
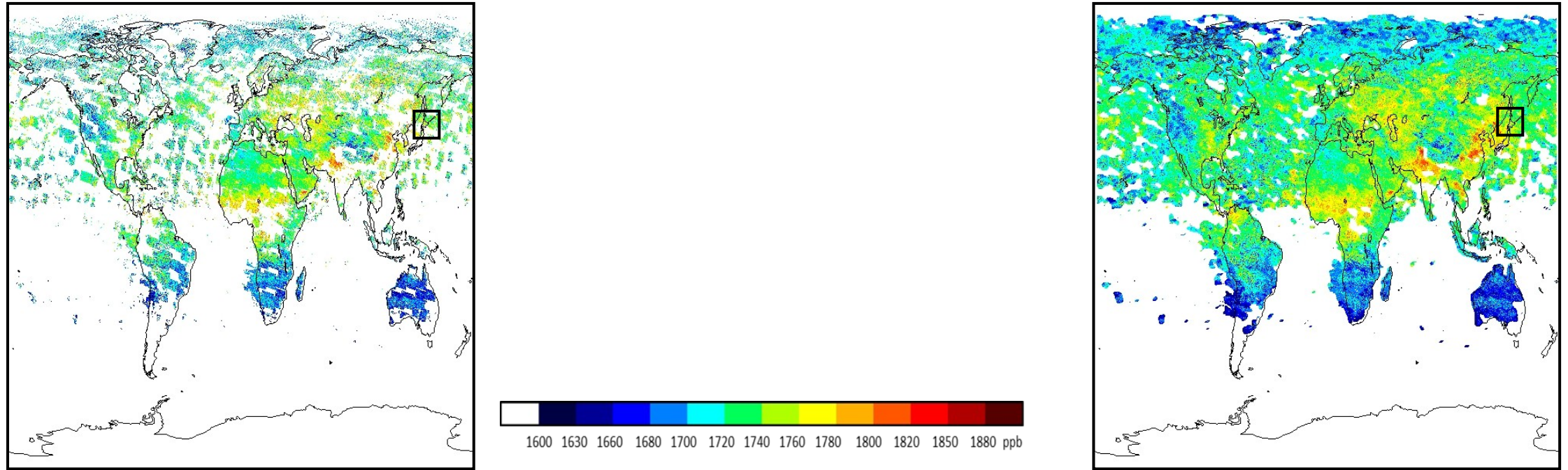
Concentration of CH₄ Emission

- Emission of CH₄ in ocean regions is extremely low in comparison with that in continental regions, and the CH₄ observed in continental regions is mainly due to **advection from land** regions.
- To simplify the air currents, we treat the CH₄ concentration in ocean regions at a given latitude as the background CH₄ concentration over continental regions at the same latitude.
- The **CH₄ emission concentration** at a location is then found by determining the **difference between the CH₄ concentration over continental regions and the background CH₄ concentration at the same latitude**.

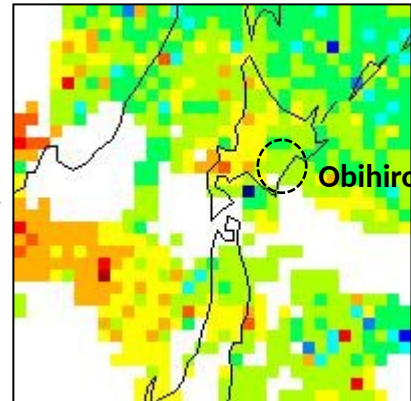


RESULT

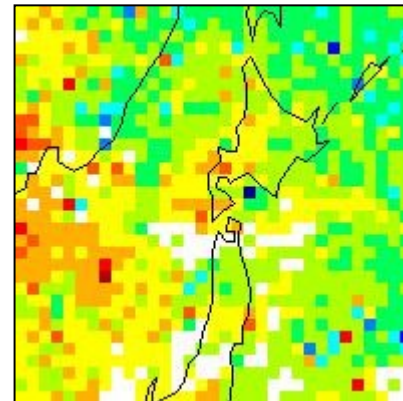
3-stage Interpolation for cloud-free image



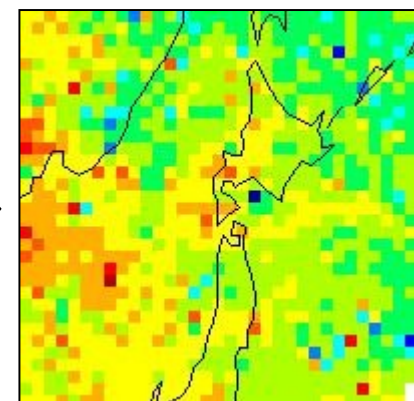
(a) 15-day composite image



(b) Spatial Interpolation

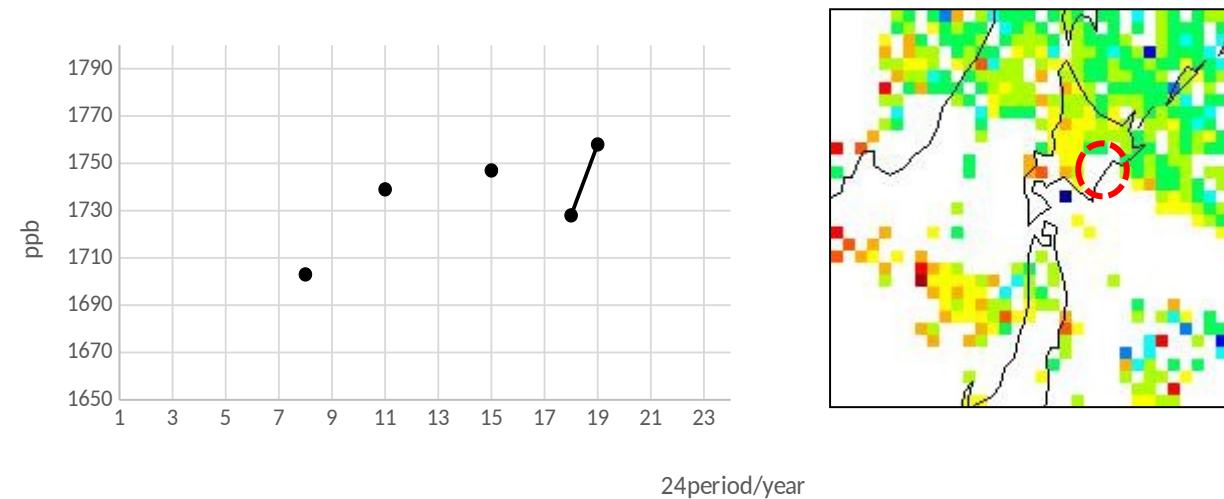


(c) Time-series Interpolation

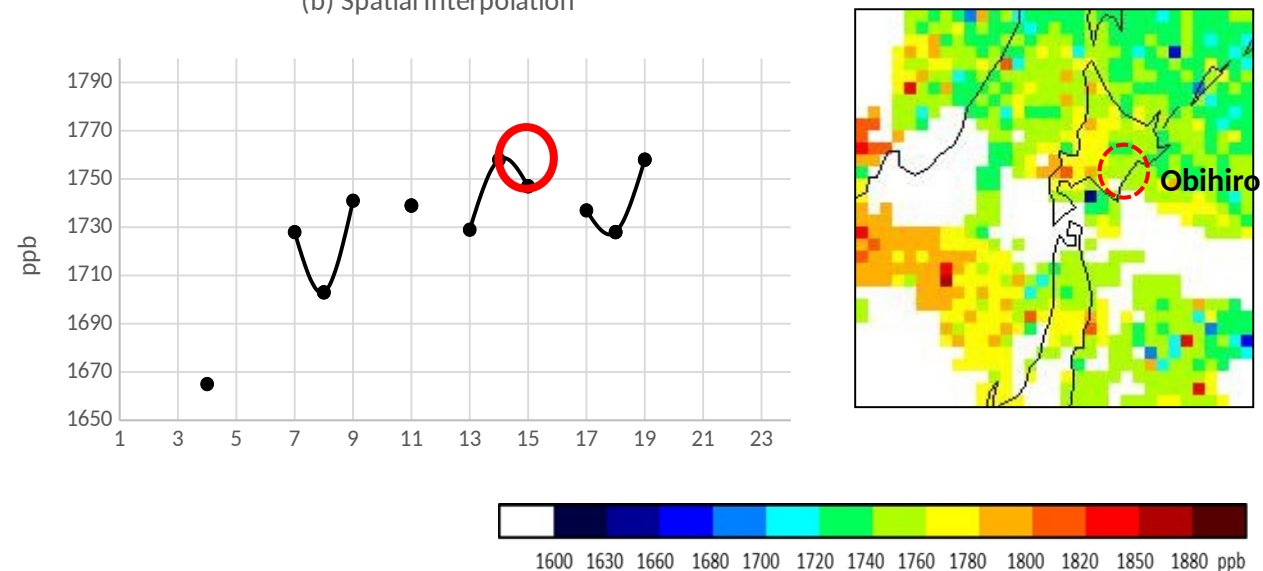


(d) Spatial Interpolation

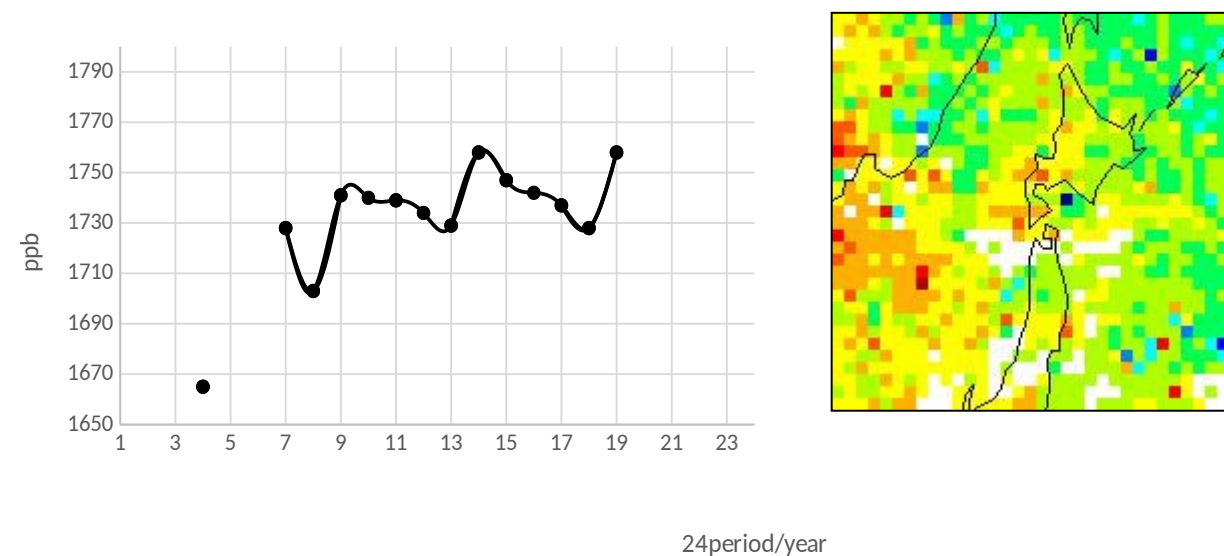
(a) 15-day composite image



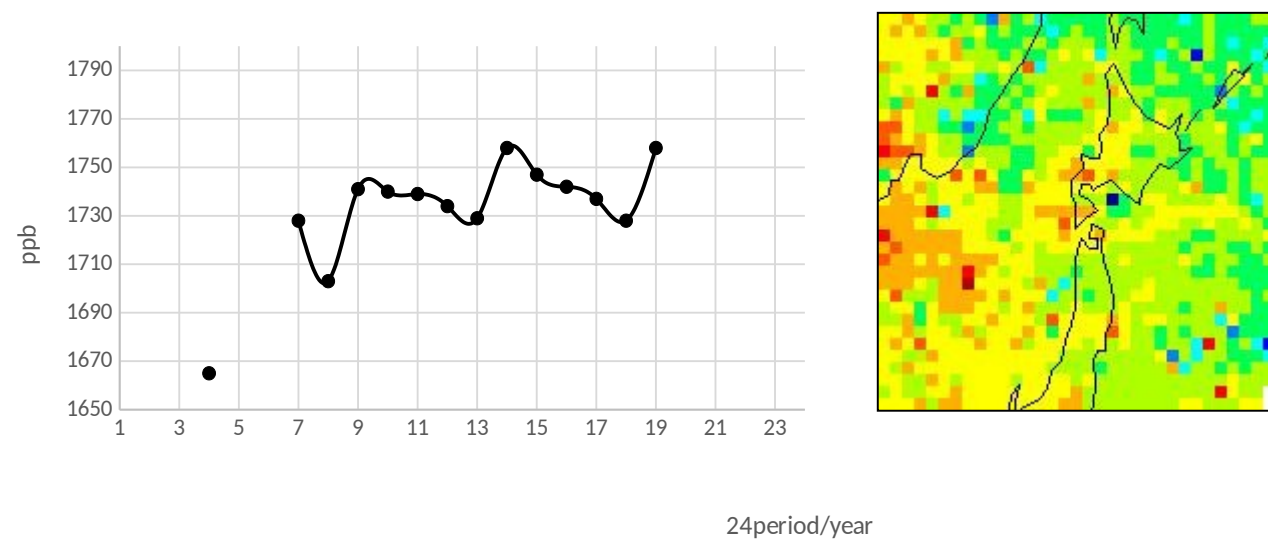
(b) Spatial Interpolation



(c) Time-series Interpolation



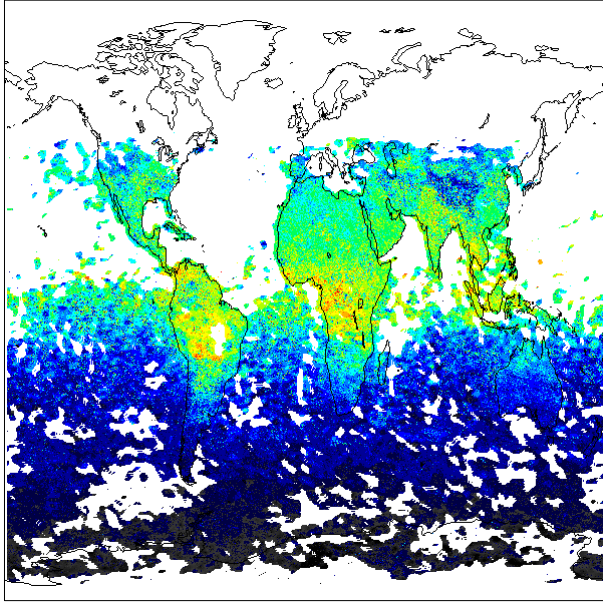
(d) Spatial Interpolation



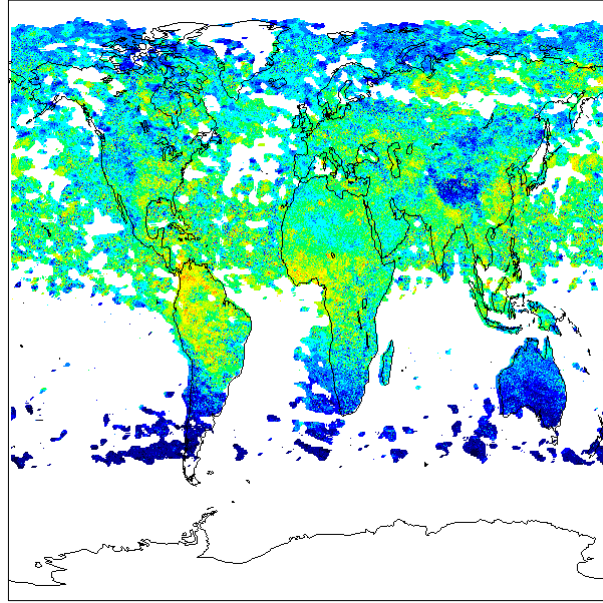
● Original data or Interpolated data

○ Interpolate data

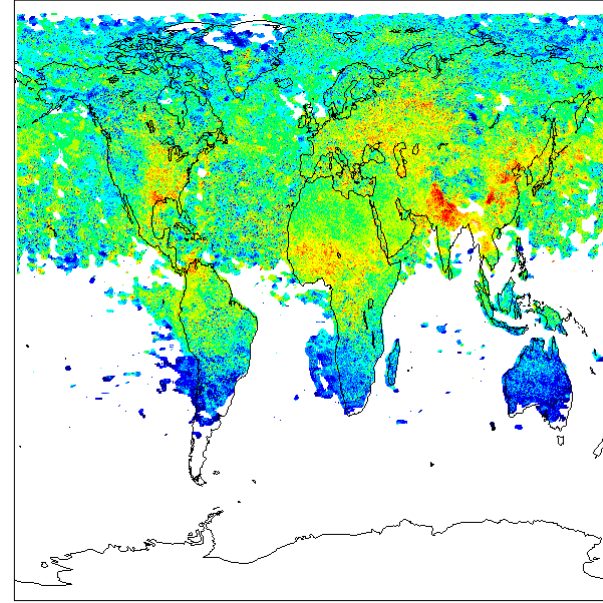
Time-series changes in methane concentration



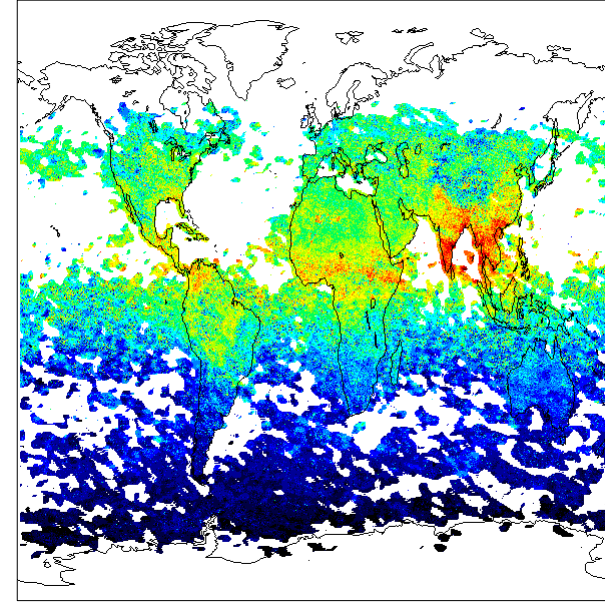
(a) January
April



(b)

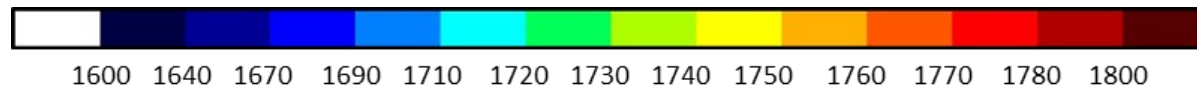


(c) July



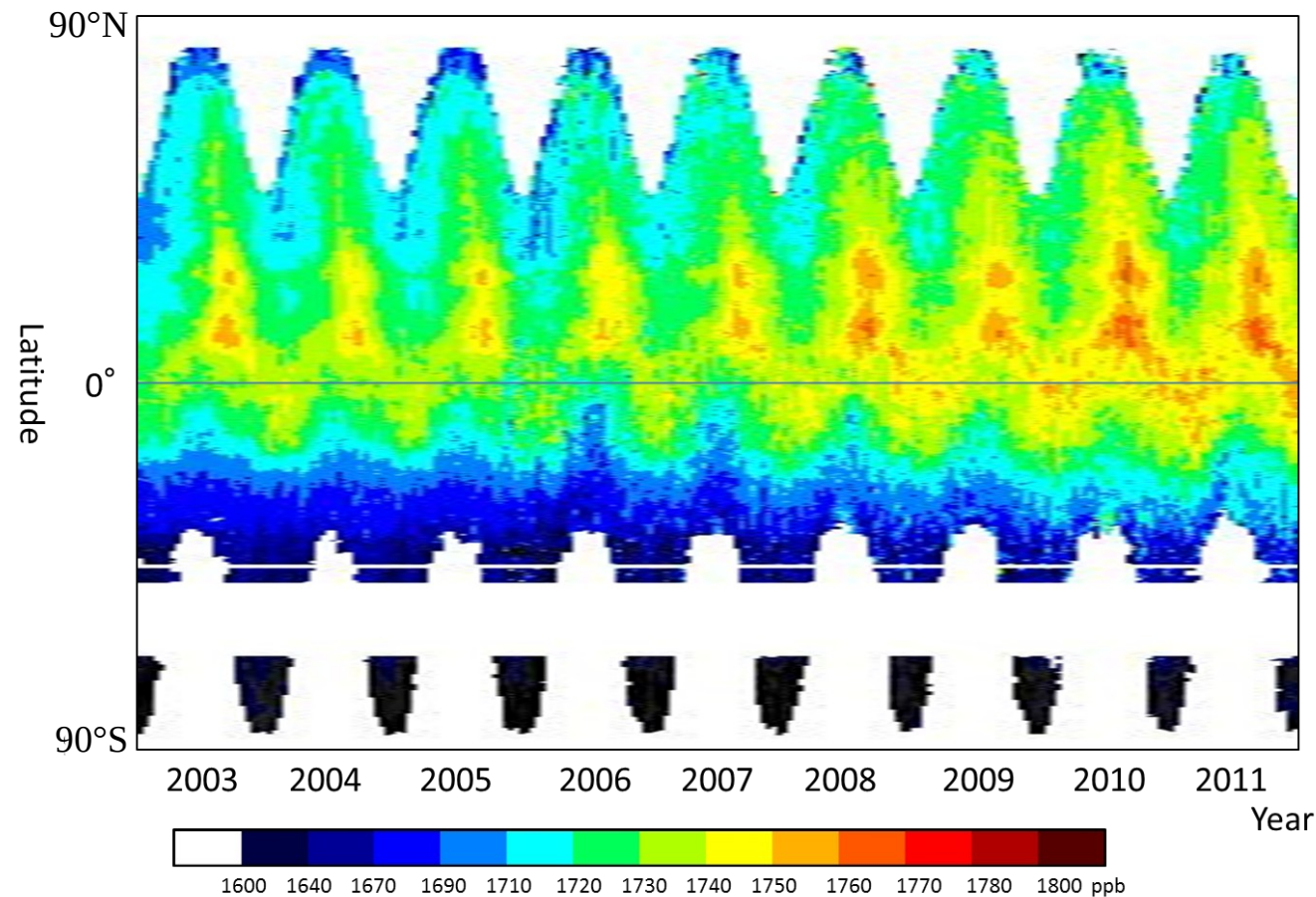
(d) October

2004

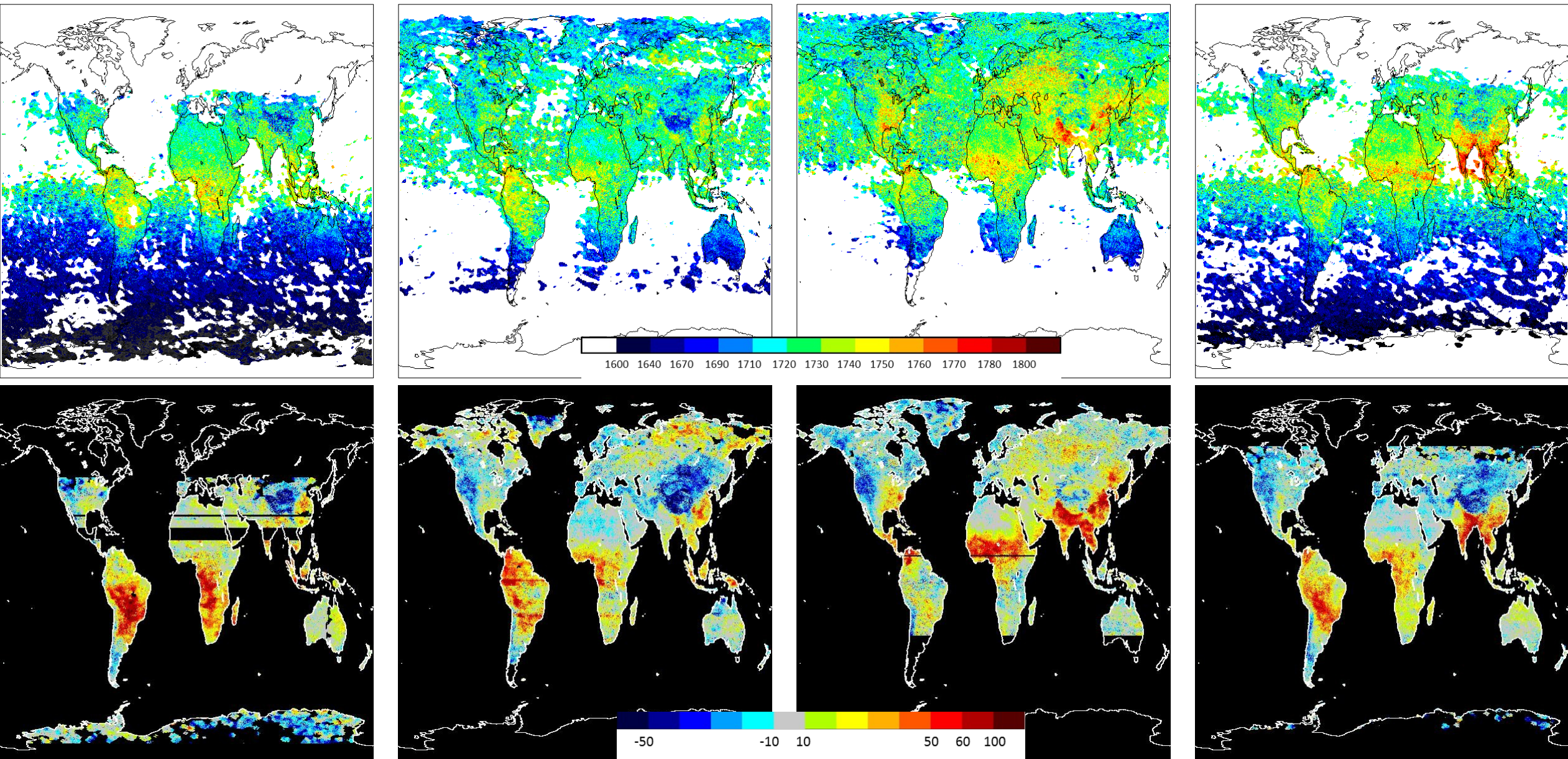


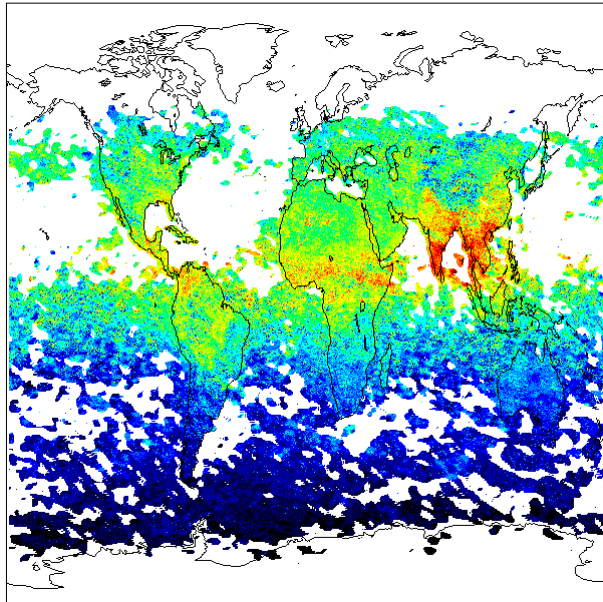
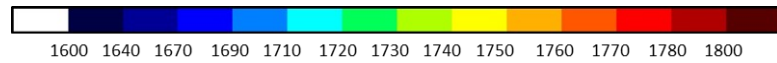
Interannual Variations in Mean CH₄ Concentration by Latitude Zone Over Continental Regions

- Figure shows the mean CH₄ concentration by latitude zone using data for only continental regions over 9 years (216 data points), from 2003 to 2011.
- Seasonal variations can be seen in the mean CH₄ concentration, with low values in spring, [increasing values from summer through autumn](#), and low values in winter.
- There are many [wet rice paddies](#) (which are a CH₄ source) located in the range of 10°N to 23°N, and the CH₄ concentration exhibits a maximum of 1850 ppb from August to October.
- It is clear that although the interannual variations in CH₄ were low from 2003 to 2006, from 2007 the concentrations increased annually, centered on the lower latitudes of the northern hemisphere, and the range of areas of higher concentration also spread.

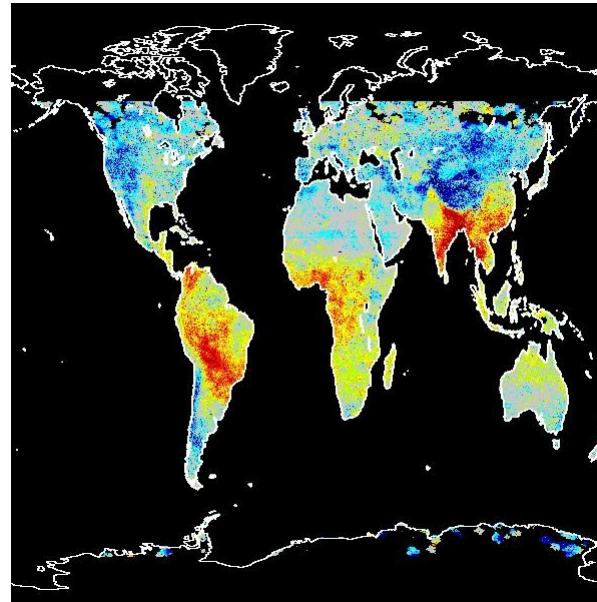
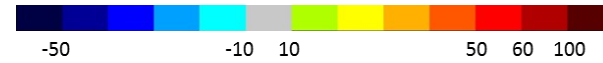


Comparison CH₄ Concentration and CH₄ Emission Concentration

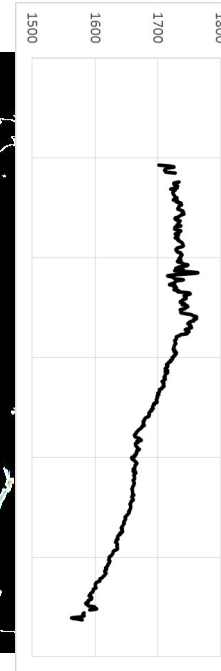




CH₄ concentration

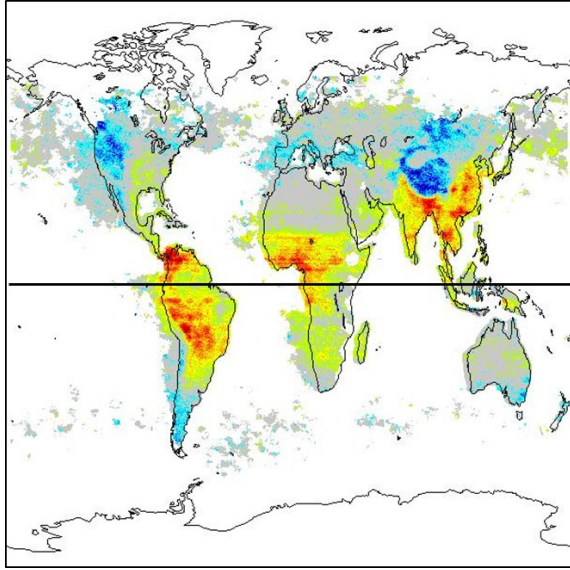


CH₄ emission concentration

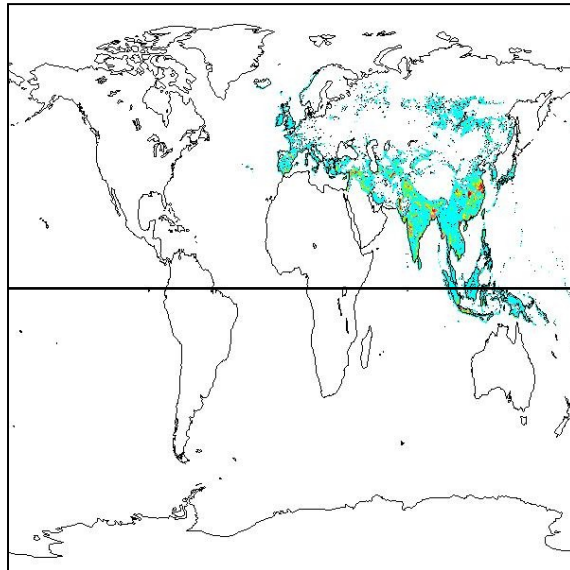


It is clear that a maximum of 100 ppb of CH₄ is emitted from the wet rice paddy region of Southeast Asia and 60 ppb or more is emitted from West Africa and central Africa in July, and a maximum of 80 ppb is emitted from South America from October to April.

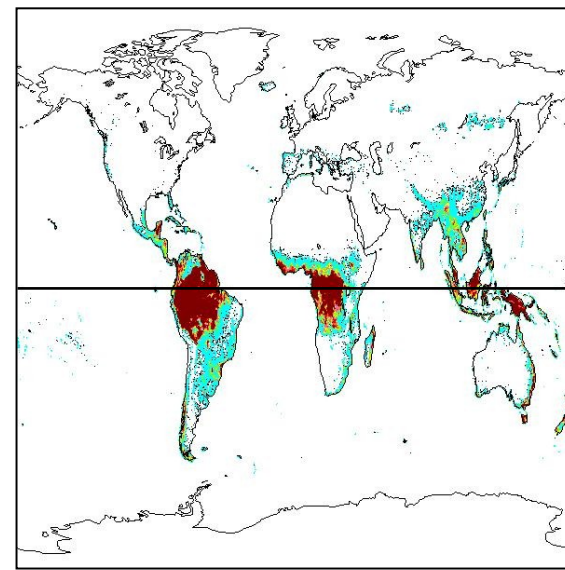
Relation between CH₄ Sources and Land Cover



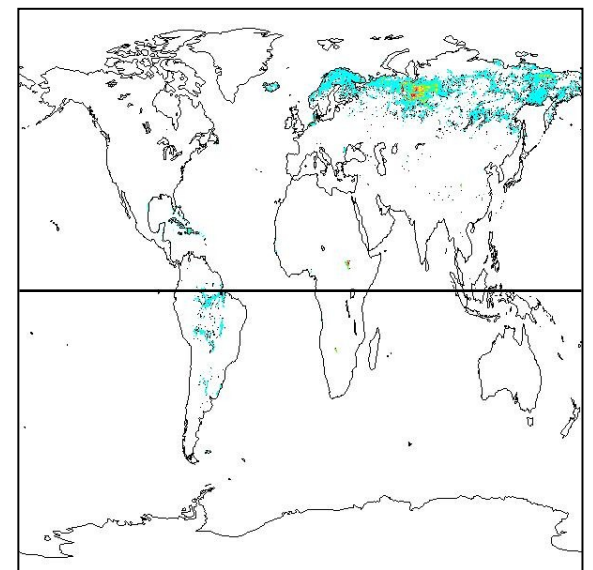
Average annual
emission concentration



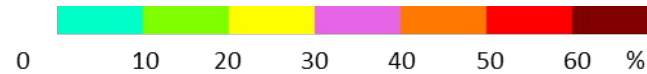
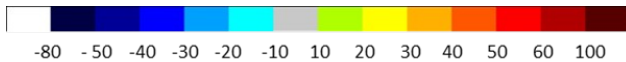
Coverage of rice paddies



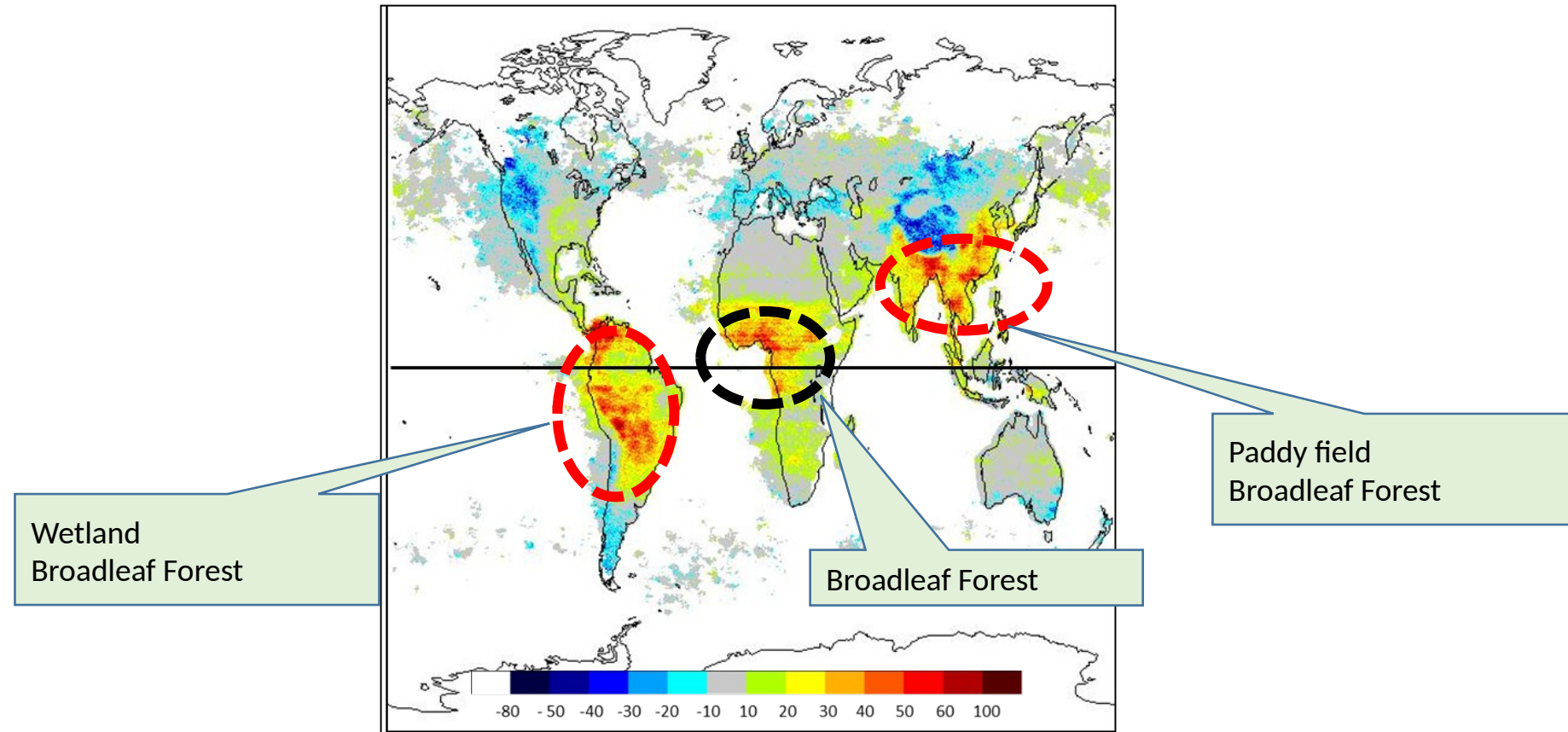
Coverage of evergreen
broad-leaf forest



Coverage of Wet land



CH₄ Emission Concentration 2009

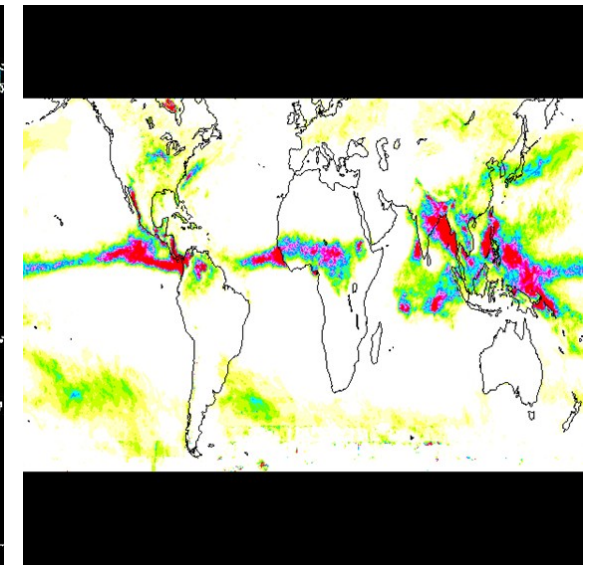
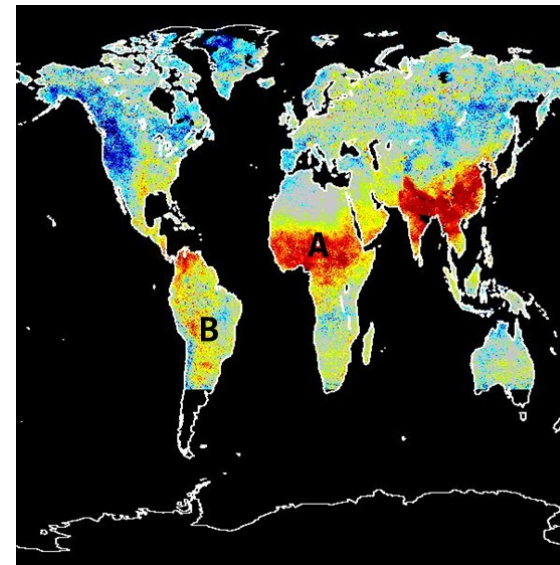
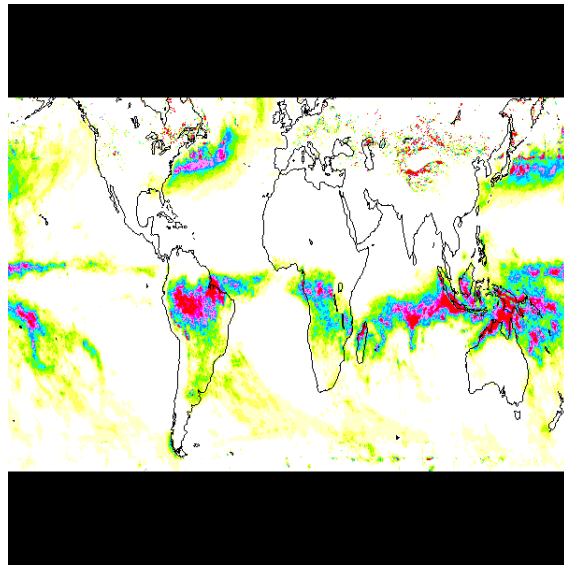
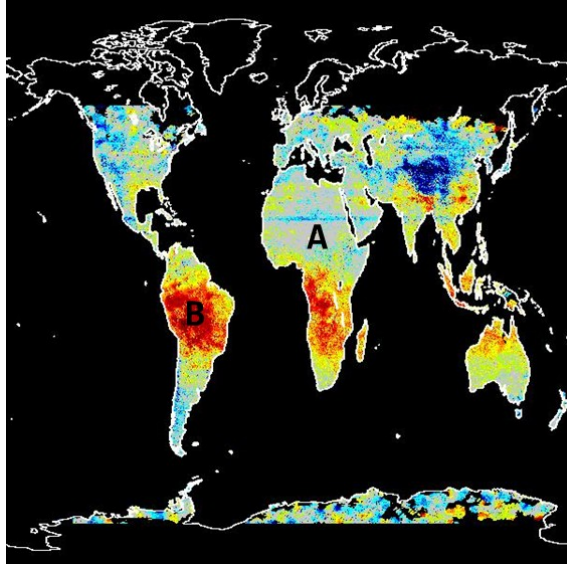


CH₄ sources are mainly paddy field in Southeast Asia and Broad leaf forest area.
Common point of the three areas is tropical area which is sufficient water and temperature.

Comparison between CH₄ Emission concentration and Precipitation

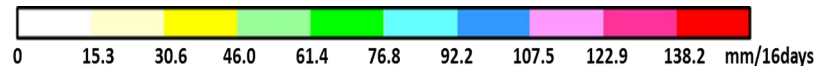
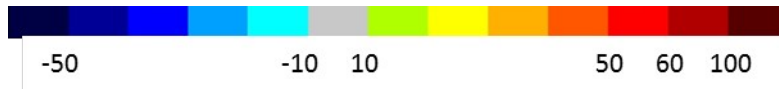
CH₄ Emission concentration

Precipitation



February

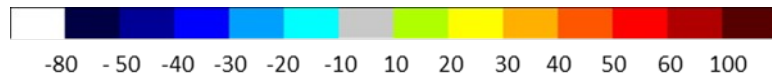
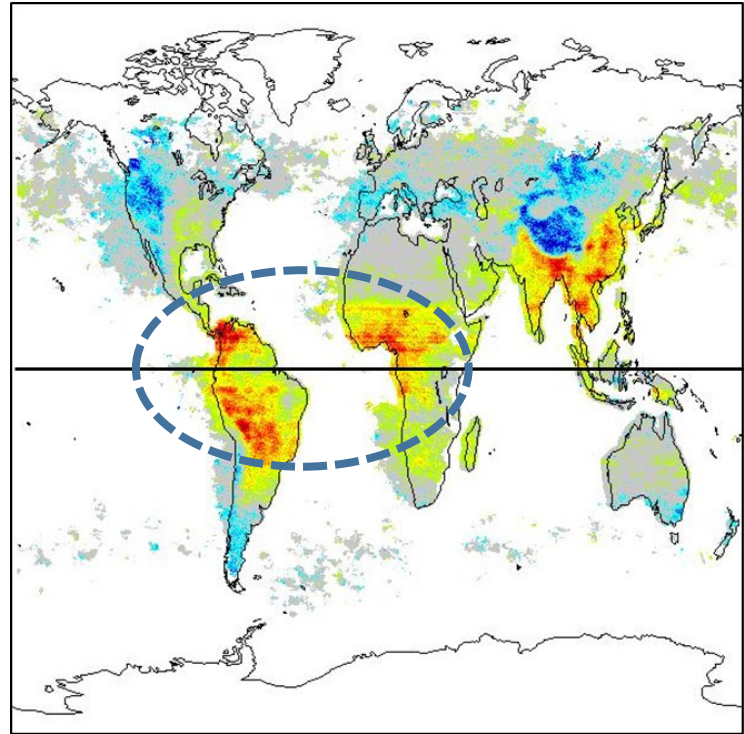
July



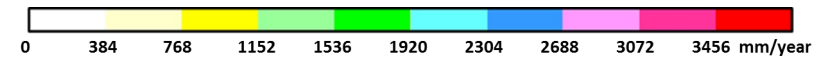
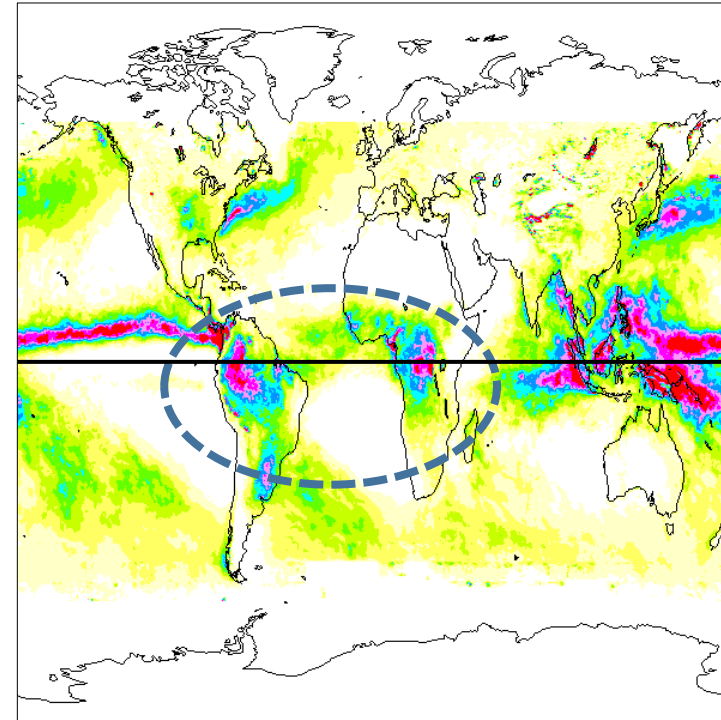
Left images show CH₄ emission concentration and right images show precipitation. Area A (near the Lake Chad, there is little rain in Feb (lower than 15mm/16days) and CH₄ emission concentration is near to zero. But in July, precipitation is higher than 122mm/16days. CH₄ emission concentration is also higher than 50ppb. Area B is South America, where the rainfall is higher than 122mm/16days and CH₄ emission concentration is also higher than 60ppb. But In July, it is lower than 15mm/16days and ch₄ emission concentration is lower than 30ppb.

Comparison between CH₄ Emission concentration and Precipitation

CH₄ Emission concentration



Precipitation



2009

The left image shows yearly CH₄ emission concentration and right image also shows yearly precipitation. It can be seen that the emission concentration is higher than 30ppb in South America and Central Africa.

The annual precipitation is more than 1900mm near the equator of 2 areas . Methane and carbon dioxide is generated in the process of decomposition of organic matter with sufficient water and temperature.

CONCLUSIONS

- When investigating CH₄ source distribution by using satellites, it is necessary to know the existing **background concentration**.
- In this research, we assumed that since CH₄ is not emitted in ocean regions, any CH₄ that does exist in the **ocean regions is due to the effect of advection from continental regions**.
- The **surface CH₄ emissions concentrations** are found from the **difference between the surface CH₄ concentration and the CH₄ concentration in ocean regions** at the same latitude.
- From the results for the annual CH₄ emissions concentrations the **wet rice paddy** regions at the lower latitudes in the northern hemisphere, which are well-known as an existing source, were confirmed to be the largest **anthropogenic source** of CH₄.
- However, we newly found that tropical evergreen broad-leaf forests in **South America and central and western Africa** are CH₄ sources, which was not previously widely known.

CONCLUSIONS

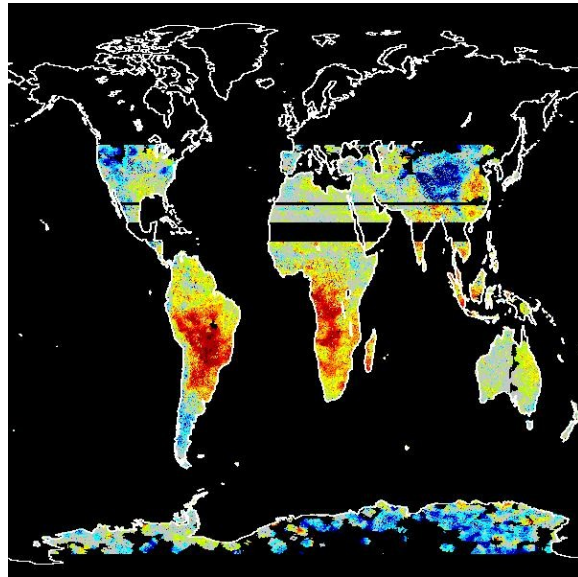
- Seasonal variation in CH₄ emission concentration in central and western Africa and South America is strongly tied to the **wet season**.
- According to the **Koppen climate classification**, these three regions have climates of **tropical rainforest, savannah, and tropical monsoon**.
- When the wet season arrives, the ground surface **temporarily changes into an anaerobic environment** due to the abundance of rain, and CH₄ is readily emitted as a result.
- CH₄ that is produced underground is **emitted** above ground from the leaves of plants by **evapotranspiration of the plant** (Megonigal and Guenther, 2008).
- Furthermore, it is thought that CH₄ that is produced near the ground surface diffuses directly into the atmosphere.

FUTURE WORK

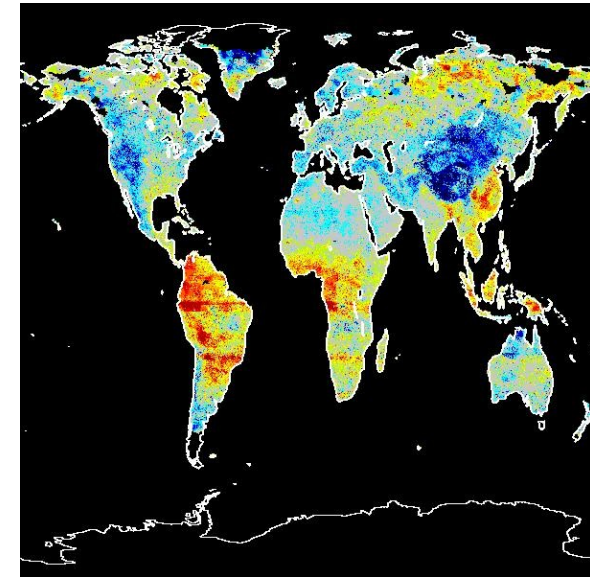
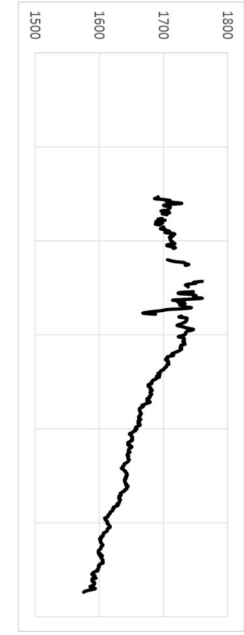
- The IPCC (2007) report estimated that much more CH₄ emissions come from [anthropogenic sources](#) than from natural sources, the results of this research showed that natural CH₄ sources are more widely distributed than anthropogenic sources.
- The extraordinary increase in CH₄ since the industrial revolution is clearly the result of anthropogenic sources.
- We need to investigate the time-series changes in the CH₄ sources that were found in this research and clarify the cause of increases in CH₄ since 2007.

Thank you for your
attention.

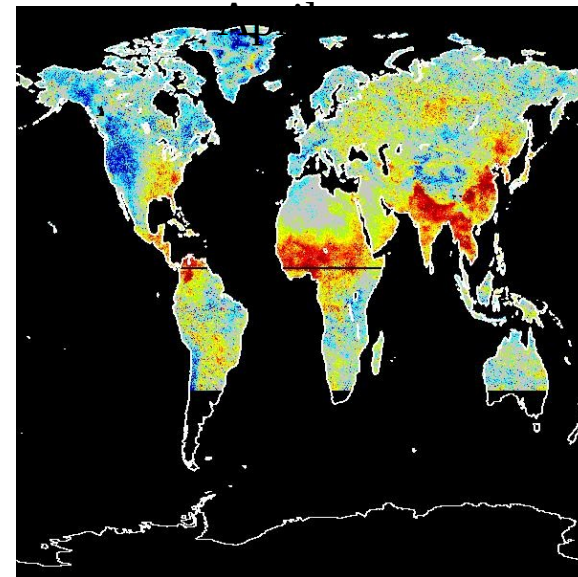
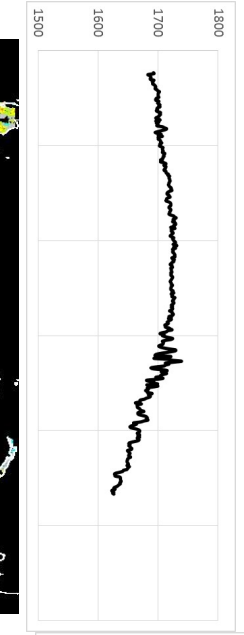
CH₄ Emission Concentration and Background concentration



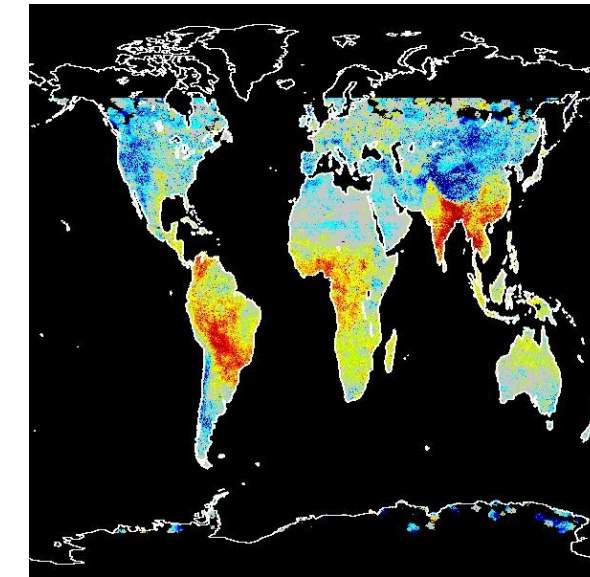
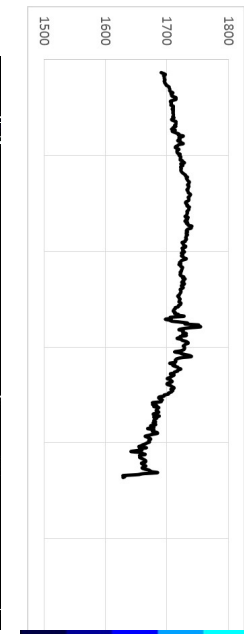
(a) January



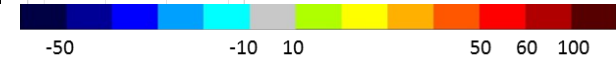
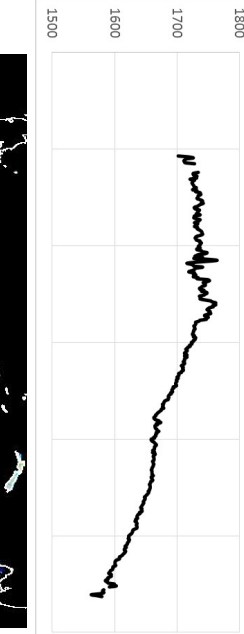
(b)



(c) July



(d) October



	CH ₄ 発生濃度	水田	畑	Open Tree	常緑広葉樹林	低木	その他
洞庭湖	61ppb	79 %	19 %				2%
四川盆地	65ppb	25 %	62 %				13%
西江周辺	61ppb	31 %	15 %	20 %			34%
ラオカイ	60ppb	22 %		50 %	11 %		17%
ナコーンサワン	63ppb	25 %	25 %	24 %			26%
チャムナ川	63ppb	53 %	30 %	12 %			5%
マガンゲ	61ppb		40 %		48 %		12%
サンタアナ	75ppb		38 %	31 %	23 %		8%
オニチャ	46ppb				57 %	29 %	14%
カメルーン	35ppb		8 %		80 %		12%

Compare background CH₄ concentration and WDCGG data

